

Honors Calculus III, Homework 2

Please upload solutions to **Question 1(a, b, c) and Question 2** by **8pm on Friday, April 7**. You can use any results you want from lectures, provided you state them clearly and you are not explicitly asked to prove them in the question, but make sure to write complete proofs. The remaining problems will not be graded, and you should not hand them in, but you are still expected to learn how to do them, unless it says “optional”. As usual, for full credit you need to show and explain your work.

1. (a) (5pt) Let $f_n : X \rightarrow \mathbf{R}$ be a sequence of bounded functions on a set X of real numbers. Assume that $f_n \rightarrow f$ uniformly on X . Prove that f is bounded.

- (b) (5pt) Prove that

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n} \tag{0.1}$$

converges pointwise to $\log(1+x)$ on $(-1, 1)$. (Hint: you could start by estimating the remainder in Taylor’s theorem for $\log(1+x)$, or by differentiating (0.1) termwise.)

- (c) (5pt) Deduce that (0.1) does not converge uniformly on $(-1, 1)$.
 - (d) More generally, assume that a power series converges pointwise to an unbounded function on a bounded interval. Prove that the convergence is not uniform.
2. (5pt) Let $F = \sum_{n \geq 0} \alpha_n x^n$ be a power series. Assume that the sequence

$$\{|\alpha_n / \alpha_{n+1}|\}$$

converges to a limit R (which can be either finite or infinite). Use the ratio test to prove that R is the radius of convergence of F .

(Hint: you are supposed to prove that the series converges if $|x| < R$ and diverges if $|x| > R$.)

3. Give examples (with proof) of power series F, G and H with radius of convergence 1 such that

- (a) F converges at $x = 1$ and $x = -1$,
- (b) G converges at $x = 1$ but not at $x = -1$,
- (c) H does not converge at $x = 1$ and does not converge at $x = -1$.

4. Spivak, *Calculus*, Chapter 24: Problems 13, 14, 25, 26.